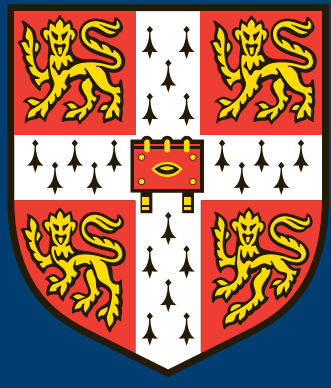


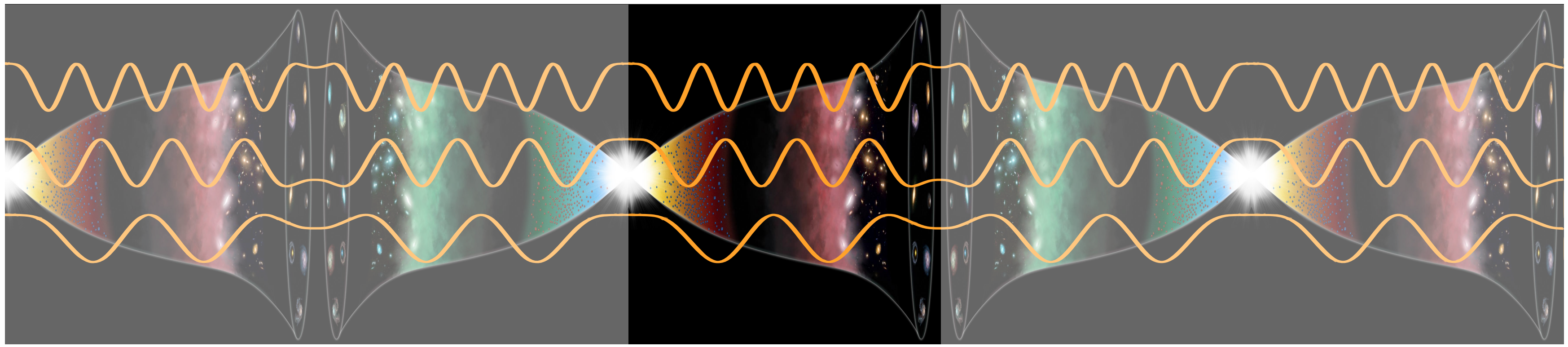


# CPT-Symmetric Kähler–Dirac Fermions

## Implication in Cosmology and Fundamental Physics

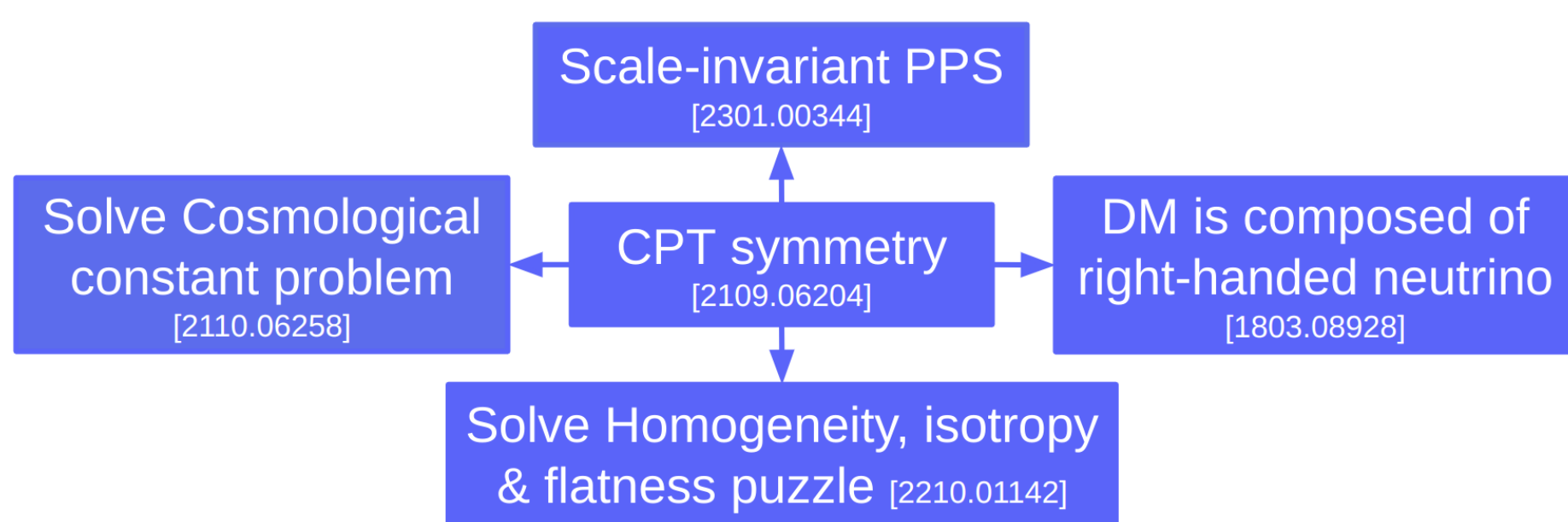
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### Introduction: CPT-Symmetric Universe

Boyle & Turok: the Big Bang acts as a **CPT mirror** → can elegantly solve lots of problems in cosmology.



### Introduction: Kähler–Dirac Fermions

**Differential-form formulation:**

$\Phi = \sum_{p=0}^4 \varphi_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}$  is a polyform with  $1 + 4 + 6 + 4 + 1 = 16$  complex components.

$$S_{KD} = \langle \bar{\Phi} | (K - m) \Phi \rangle, \quad (K - m) \Phi = 0, \quad K = d - d^\dagger, \quad -K^2 = \square.$$

**Spinor formulation:** replacing  $dx^\mu \rightarrow \gamma^\mu$  assembles the 16 components into a  $4 \times 4$  matrix  $\Psi$ :

$$(i\gamma^\mu \partial_\mu - m)\Psi = 0, \quad S_{KD} = \int d^4x \text{Tr}[\bar{\Psi}(i\not{\partial} - m)\Psi], \quad \bar{\Psi} \equiv \gamma^0 \Psi^\dagger \gamma^0.$$

Each column of  $\Psi$  satisfies the ordinary Dirac equation. The symmetry group is  $\text{Spin}(3, 1)_{\text{Lorentz}} \times U(2, 2)_{\text{internal}}$ .

### Negative-Norm States & Two-Sheeted Spacetime

**Problem:** The extra  $\gamma^0$  on the left of  $\bar{\Psi}$  means half the columns have the wrong-sign Lagrangian. Diagonalising  $\gamma^0 = \Omega \Lambda \Omega^T$ , with  $\Lambda = \text{diag}(+1, +1, -1, -1)$ :

$$S_{KD} = \int d^4x \text{Tr}[\Lambda \Psi'^\dagger \gamma^0 (i\not{\partial} - m) \Psi'], \quad \Psi' \equiv \Psi \Omega.$$

Columns 3–4 of  $\Psi'$  have the wrong sign  $\Rightarrow$  **negative-norm states (ghosts)!**

**Resolution [Donoghue & Menezes, 2019]:** a wrong-sign action = reversal of the causality arrow =  $PT$  reversal. The two pairs of columns live on **two sheets of spacetime**,  $PT$ -mirrors of each other; both sheets have positive energy and norm.

### KD-Majorana condition & SM Unification

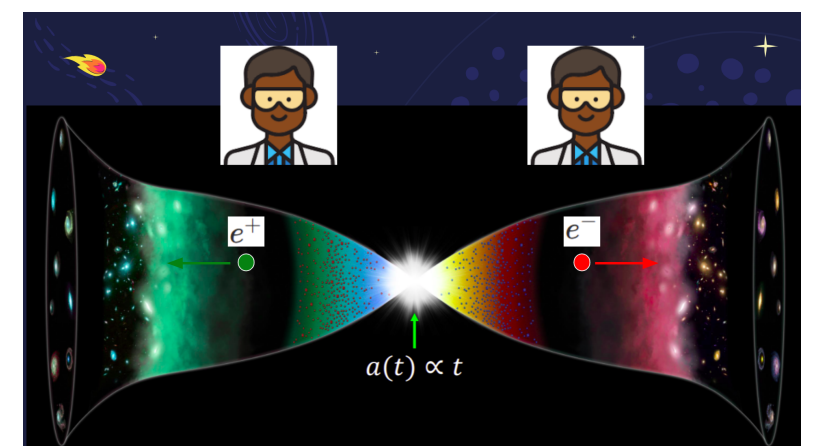
**KD-Majorana condition:** to avoid coupling between two sheets and doubling of the particle content, impose

$$\Psi^c \equiv (i\gamma^2)\Psi^*(i\gamma^2) = \Psi.$$

This forces the four columns to take the form:

$$\Psi' = (\psi_1, \psi_2, -\psi_2^c, \psi_1^c).$$

Every LH particle on the forward sheet is *always* paired with a RH anti-particle on the backward sheet – a manifestation of CPT symmetry.



**SM fermion content in KD language:** one SM generation (including  $\nu_R$ ) = 4 KD-Majorana fields  $\Psi'_{A=0,1,2,3}$  (leptons + three quark colours). All SM fermion kinetic & Yukawa terms unify:

$$\mathcal{L}_{KD} = \text{Tr}[\bar{\Psi}_A(i\not{\partial})\Psi_A], \quad \mathcal{L}_Y = \text{Tr}[\mathcal{H}^T \hat{\Psi}'_{A,L} \Psi'_{A,R} \mathcal{Y}_A^{ij}] + \text{h.c.}$$

Gauging  $SU(3) \subset SU(4)$  and  $SU(2) \times U(1) \subset U(2, 2)$  reproduces *all* SM fermion gauge interactions.  $\nu_R$  is required by the geometry of the KD field.

### Physical Implications

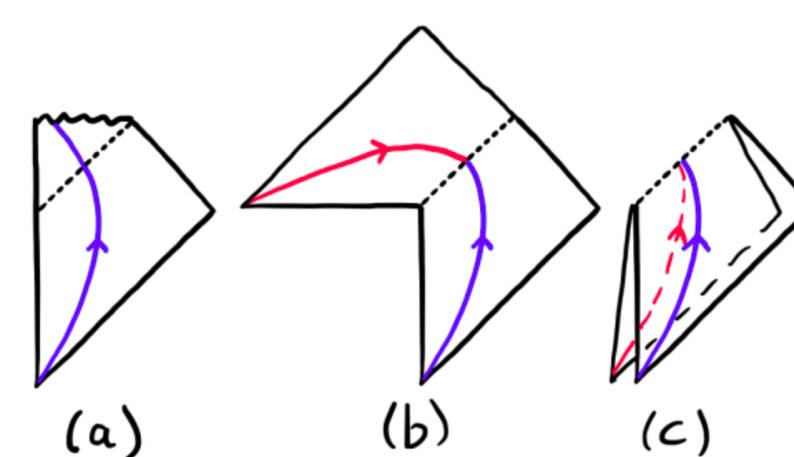
(1) Theoretical Foundation of the CPT Mirror

KD-Maj. field theory  $\leftrightarrow$  CPT mirror at Bang cosmology

(2) Black Mirror

[arXiv:2412.09558]

- The BH interior does not exist
- The horizon mirrors our exterior to the “other” exterior (the anti-universe)
- Infalling particles *annihilate* with anti-partners from the anti-universe at the horizon
- The KD-Majorana condition explains why every infalling particles are paired



(3) Description of *Tenet*?!



Same as CPT-sym. uni.:

1. There is anti-universe with time goes backward
2. The transition gates are **black holes**

**Difference from Tenet:**

1. You and anti-you **annihilate** at the transition gate (horizon)
2. Viewed from our universe: you emerge from a **white hole** and find time reversed –but you *cannot change anything*, only re-experience the past

### Future work

1. Construct KD Chiral gauge theory by considering two sheeted spacetime [Vaibhav, Latham (preparation)]
2. String theory in 4D: consider KD fermions in superstring might form a ghost free theory without extra dimensions.